Jason Nguyen

+1(315)975-8809 | nguyenjanh01@gmail.com

https://www.linkedin.com/in/jason-nguyen-360b7b275

https://website-doranis.vercel.app/

CS graduate seeking a software or data engineering role, with research and internship experience spanning ML pipelines, distributed data systems, embedded systems, and full-stack development.

EDUCATION

SUNY University at Buffalo August 2021 - May 2026

Bachelor of Science, Computer Science

Relevant Coursework: *Systems Programming, Data Structures, Algorithms and Complexity, Discrete Structures, Software Engineering Concepts, Computer Security, Distributed Systems, Reinforcement Learning, Intro Machine Learning, Computer Vision & Image Processing, Applied Human & Computer Interaction, Algorithms Analysis & Design, Linear Algebra, Calculus 1, 2, & 3, Data Intensive Computing*

Cicero-North Syracuse High School September 2017 - June 2021

Graduated with Honors

EXPERIENCE

ARISE Adaptive Design May 2024 - August 2024

Adaptive Design Intern

- Engineered and validated object-oriented firmware in a Linux environment for 2 embedded systems using Arduino microcontrollers, writing unit tests in C++ that verified sensor input thresholds, signal timing, and hardware-software integration across 23 test cases.
- Reduced latency by 15% based on serial timing logs before and after the firmware change, through iterative signal testing and firmware optimization, using serial debug logs to isolate and resolve bottlenecks.
- Authored hardware configuration documentation and debugging reference guides standardizing setup procedures across 3 device variants, cutting onboarding and re-configuration time for the team.

Randstad February 2023 - April 2024

Data Analyst Intern

- Cleaned and validated datasets of 9,000 records using Excel (pivot tables, VLOOKUP, conditional logic) and SQL queries, identifying and resolving 4 categories of recurring data entry errors across reporting pipelines.
- Built 2 standardized reporting templates that automated previously manual aggregation steps, reducing report turnaround time by 1 hour per cycle.
- Audited data entry workflows using root cause analysis to trace discrepancies to upstream process gaps, producing a findings report that was adopted to reduce error rates in weekly deliverables.

United Radio January 2020 - September 2020

Electronics Technician

- Diagnosed and repaired consumer electronics at a rate of 750 devices per week, performing circuit board-level fault analysis using multimeters to trace electrical failures and isolate faulty components.
- Executed component-level soldering repairs using both schematics/repair manuals and hands-on diagnostic techniques, restoring devices across a high-volume repair pipeline.
- Generated repair reports documenting fault findings, components replaced, and resolution outcomes, maintaining accurate service records across all processed devices.

SKILLS

ML/AI: PyTorch, TensorFlow, PySpark, OpenCV, NumPy, Pandas

Web/Full-Stack: React, Svelte, TypeScript, JavaScript, HTML/CSS, Docker, Node

Languages: Python, Rust, Java, C/C++, C#, Go, Scala, GDScript

Tools & Platforms: Git, Bash, MySQL, SQLite, PostgreSQL, MongoDB, Google Cloud, Arduino, Figma, JIRA

RESEARCH

Large-Scale Spotify Track Popularity Analysis with PySpark & ML | Link to Report

- Designed and executed a large-scale data analysis and machine learning pipeline on 97,624 Spotify tracks using PySpark on HDFS. Built a Random Forest Regressor to predict song popularity from audio features; improved R^2 by 41% (0.17 to 0.24) through hyperparameter tuning via 3-fold cross-validation. Performed genre popularity distribution analysis across 114 genres and audio feature significance testing (Welch's t-test, Cohen's d) across six popularity tiers, identifying acousticness and danceability as the strongest predictors of viral performance.

Handwriting to Text: CRNN + LLM Study Guide Pipeline | Link to Report

- Built an end-to-end pipeline that converts handwritten student notes into structured study guides. Trained a Convolutional Recurrent Neural Network (CRNN) with CTC loss on 400k+ word images from the IAM Handwriting Database and Kaggle, achieving 12.47% Character Error Rate. Integrated Llama 3.3 70B via the Groq API for downstream summarization, reaching ROUGE-1 F1 of 0.423. Evaluated against Tesseract and BLIP-2 baselines.

PROJECTS

Personal Portfolio Website | *React, JavaScript, CSS, HTML, Git, Vercel* | https://website-doranis.vercel.app/

- Built and deployed a personal portfolio website using React and Vercel, implementing a responsive single-page application with custom CSS styling and component-based architecture, serving as a centralized showcase of projects and professional background.

BDO Loot Tracker | *Python, Svelte, CSS, SQLite, Git* | https://github.com/janhnguyen/BDO-Loot-Tracker

- Built a desktop application using Python, Svelte, SQLite, Git, and packaged distribution tools with a modular architecture integrating UI, local data persistence, and data processing components to track and analyze in-game item data.

Lost Ark Logs | *Rust, Svelte, TypeScript, SQLite, Git* | https://github.com/snoww/loa-logs

- Designed and implemented UI/UX components for a Rust-based gameplay analytics tool, optimizing SQLite schema and queries to improve data storage and retrieval performance. Prioritized user feedback to develop product features. Owned feature development end-to-end within a multi-developer Git workflow, driving code reviews, issue tracking, and iterative releases.

Tic-Tac-Toe Bot | *Python, PyTorch* **|** https://github.com/janhnguyen/rl446tictactoe

- Developed a reinforcement learning framework in Python using PyTorch, implementing DQN, PPO, SAC, and Rainbow agents with a custom Tic-Tac-Toe environment, reward functions, and training pipeline to evaluate agent performance and learning efficiency.

OCR Party Inspection | *Python, PyTesseract, OpenCV, Numpy* **|** https://github.com/janhnguyen/partyInspection

- Built a Python application using OpenCV and PyTesseract to capture and process game client screenshots, extract structured data via OCR and image recognition, and transmit results to an external web service through automated data pipelines.

Social Media Site | *React, CSS, JavaScript, HTML, Git, Docker* **|** https://github.com/janhnguyen/djibouti

- Developed a full-stack social media web application using React and a custom backend API, implementing responsive UI components, RESTful client-server communication, and containerized deployment with Docker to ensure scalability, portability, and maintainable architecture.

LEADERSHIP & COMMUNITY INVOLVEMENT

Lost Ark Nexus Contributor | https://www.lostark.nexus/ | https://shorturl.at/18Kle

- Authored and maintained a comprehensive technical guide analyzing system mechanics, optimization strategies, and performance tradeoffs for a large online gaming community.
- Interpreted gameplay metrics and iterative updates to refine recommendations. Actively supported users through Q&A and technical review.

UB Open Source Club

- Collaborated as an open source club member to develop indie games, contributing to coding, design, and testing, which enhanced project quality and innovation. Participated in brainstorming sessions and provided feedback.